Current fluctuations in the Toda lattice

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based on work with Amol Aggarwal

What this talk is about

- Toda lattice is a deterministic, *Hamiltonian system* in infinite volume
- Random initial data, *thermal equilibrium*
- Main results: Gaussian fluctuations of integrated currents
- Proof strategy and tools: quasi-particles and localization

Toda lattice

Toda lattice: historical background

19th-century origin: solitary waves. In the 1830s, John Scott Russell (a naval architect) observed a localized water wave in a canal that propagated over long distances without changing shape (the "wave of translation").

Continuum theory: KdV. In the 1960s, this phenomenon was modeled by the Korteweg–de Vries (KdV) equation, which admits *soliton* solutions: traveling waves that maintain their shape, and interact in a certain explicit way.

Toda's question (1960s). Morikazu Toda asked whether an analogous phenomenon could exist in a *discrete mechanical system*:

- particles on a line,
- nearest-neighbor interactions,
- nonlinear but still exhibits a large collection of exact solutions.

Toda lattice: Hamiltonian formulation

Phase space: One momentum and position for each $j \in \mathbb{Z}$

$$\mathcal{P} = (\mathbb{R}^2)^{\mathbb{Z}}, \qquad \{(p_j, q_j)_{j\in\mathbb{Z}}\}.$$

Equations of motion of the Toda lattice:

$$\dot{q}_j = \frac{d}{dt} q_j = \frac{\partial H}{\partial p_j} = p_j, \qquad \dot{p}_j = \frac{d}{dt} p_j = -\frac{\partial H}{\partial q_j} = e^{q_{j-1}-q_j} - e^{q_j - q_{j+1}}.$$

Classical mechanics system corresponding to the Hamiltonian:

$$H(\mathbf{p}, \mathbf{q}) = \sum_{j\in\mathbb{Z}} \left(\frac{p_j^2}{2} + e^{q_j - q_{j+1}} \right).$$

Interpretation: particles on a line with exponential nearest-neighbor interaction.

Flaschka variables and random initial data

Flaschka variables:

$$a_j = e^{\frac{1}{2}(q_j - q_{j+1})}, \qquad b_j = p_j.$$

Thermal equilibrium: For initial data, let $q_0(0)$. Then, for some $\beta, \theta > 0$, sample $a_j, b_j, j \in \mathbb{Z}$ independently, from chi and normal distributions, respectively

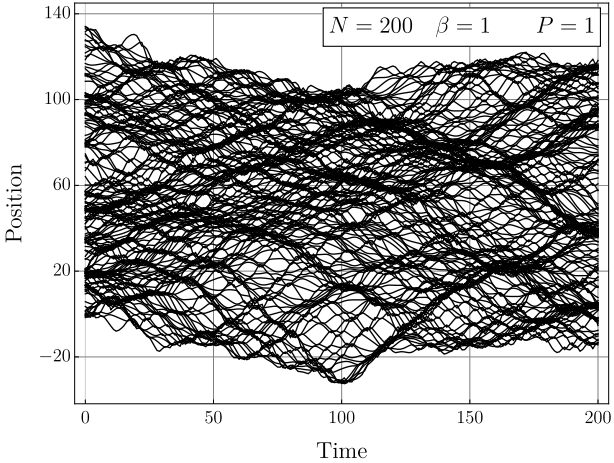
$$a_j \sim const. \left(\mathbb{1}_{a_j \geq 0} a_j^{2\theta - 1} e^{-\beta a_j^2} \, da_j \right) \tag{1}$$

$$b_j \sim const. \left(e^{-\beta b_j^2/2} \, db_j \right). \tag{2}$$

This distribution is *invariant* under the dynamics.

Goal: Analyze certain observables after a large time.

A simulation under thermal equilibrium



We thank Alberto Brollo for this simulation

Statement of first results

A first result: fluctuations of an integrated current

Integrated particle current across 0: For $t \geq 0$, define

$$J_t^{[0]} := \sum_{j \in \mathbb{Z}} -\mathbb{1}_{q_j(t)<0<q_j(0)} + \mathbb{1}_{q_j(0)<0<q_j(t)}.$$

"The net, or signed, number of particles crossing 0 in time t".

Theorem (Aggarwal-N.,2026+)

Let $\beta > 0$, and let $\theta > 0$ be small enough. For the Toda lattice initialized at thermal equilibirum,

$$\frac{J_t^{[0]} - \mathbb{E}[J_t^{[0]}]}{T^{1/2}} \xrightarrow{d} \textit{(Explicit constant)} \times \textit{(Brownian motion)} \tag{3}$$

Generalization of first result: a family of integrated currents

Spatial integrated currents: For $q, q' \in \mathbb{R}$, define

$$J_t^{[0]}(q, q') := -\sum_{q_j(t)<q'} \mathbb{1}_{q<q_j(0)} + \sum_{q_j(0)<q} \mathbb{1}_{q'<q_j(t)}$$

"The net, or signed, number of particles crossing the ray from $(q, 0)$ to (q', t) in time t.

Extension of the previous theorem [Aggarwal-N., 2026+]:

$$\frac{J_t^{[0]}(q, q') - \mathbb{E}[J_t^{[0]}(q, q')]}{T^{1/2}} \xrightarrow{d} \text{Explicit Gaussian process in } q, q', t \tag{4}$$

Two-point function

Stretch and momentum: Define the stretch $r_j(t) := q_{j+1}(t) - q_j(t)$, and recall the momentum $p_j(t)$ of particle j.

Space–time covariances: For integers $m, n \geq 0$, define

$$S_{0,0}(j,t) := \mathrm{Cov}\big(r_j(t),\, r_0(0)\big), \tag{5}$$

$$S_{1,0}(j,t) := \mathrm{Cov}\big(p_j(t),\, r_0(0)\big) \qquad S_{1,1}(j,t) := \mathrm{Cov}\big(p_j(t),\, p_0(0)\big). \tag{6}$$

Measure of "speed of mixing" in different directions.

A two-point function result: For $0 \leq m, n \leq 1$

$$S_{m,n}(j,t) \sim const(j/t) \cdot \frac{1}{t}$$

(Two-point function decays as $1/t$.)

Two-point function result

Theorem (Aggarwal-N., 2026)

Under thermal equilibrium, with $\theta > 0$ small enough, if $j = \lfloor xT \rfloor$, as $T \to \infty$ we have the asymptotic

$$S_{0,0}(j,T) \sim \alpha^2 \frac{1}{T} \langle \varsigma_0^{\mathrm{dr}}, \delta(x - \alpha^{-1} v_{\mathrm{eff}}(\lambda)) \cdot \varsigma_0^{\mathrm{dr}} \rangle_{\varrho} \tag{7}$$

$$S_{1,0}(j,T) \sim -\alpha \frac{1}{T} \langle \mathbf{F}\varsigma_1, \delta(x - \alpha^{-1}\ v_{\mathrm{eff}}(\lambda)) \cdot \varsigma_0^{\mathrm{dr}} \rangle_{\varrho} \tag{8}$$

$$S_{1,1}(j,T) \sim \frac{1}{T} \langle \mathbf{F}\varsigma_1, \delta(x - \alpha^{-1} v_{\mathrm{eff}}(\lambda)) \cdot \mathbf{F}\varsigma_1 \rangle_{\varrho} \tag{9}$$

in the sense of pairing against smooth, compactly supported functions.

These formulas were predicted by [Spohn, 2020]. Above, $\langle \cdot, \cdot \rangle_{\varrho}$ is the inner product on a certain Hilbert space $\mathcal{H}$ of functions, $\mathbf{F}$ is an operator on that space, $\varsigma_n(\lambda) = \lambda^n$ are monomial functions, and $v_{\mathrm{eff}} \in \mathcal{H}$ is not yet defined.

Comparison to ASEP: integrated current

Dyanmics: Particles on $\mathbb{Z}$ hop to the left at rate $L > 0$, and to the right at rate $L + 1$, subject to *exclusion constraint*.

Hydrodynamic limit: At large scales the *density* $\rho(x, t)$ is the *only locally conserved quantity*, and it satisfies [Andjel-Vares, 1987]

$$\rho_t + (1 - 2\rho)\rho_x = 0.$$

Integrated current: Define $J_t(q, q')$ as the net number of particles crossing the ray from $(q, 0)$ to (q', t) in time t.

KPZ scaling exponents and fluctuations: Suppose ρ is constant (e.g. product Bernoulli invariant measure). Along the characteristic direction, say $t = T, q = 0, q' = (1 - 2\rho)T$,

$$J_t(q, q') - \mathbb{E}[J_t(q, q')] \sim T^{1/3} \times (\text{KPZ-like random variable}) \tag{10}$$

Comparison to ASEP: two-point function

Two-point function: Let $\eta(t) = \{\eta_j(t)\}_j \in \{0,1\}^{\mathbb{Z}}$ be the configuration at time t. For $j \in \mathbb{Z}$, define

$$S(j,t) \sim \mathrm{Cov}\left(\eta_0(0), \eta_j(t)\right).$$

Then it is known that if $j \approx (1-2\rho)T$ (along the characteristic direction)

$$S(j,t) \sim const. \frac{1}{T^{2/3}}$$

and along other directions, $S(j,t)$ decays exponentially in T.

ASEP vs Toda: current and two-point scaling at a glance

	ASEP (KPZ class)	**Toda**
Integrated current	**Exponent:** $t^{1/3}$ **Limit:** Baik-Rains / KPZ-like (along characteristics)	**Exponent:** $t^{1/2}$ **Limit:** Gaussian
Two-point function	**Width:** $t^{2/3}$ **Size:** $t^{-2/3}$	**Width:** t **Size:** t^{-1}

Conclusion: ASEP at equilibrium exhibits different universal scaling exponents and fluctuations than the Toda lattice at equilibrium.

Characteristics in Toda: In the Toda lattice *every direction* is a characteristic direction (corresponding to some conserved quantity).

Statement of general results

In a general system of classical mechanics

Recall ASEP hydrodynamics: At the large scale, it suffices to keep track of the density $\rho(x,t)$, which satisfies the Burgers equation $\rho_t + (1-2\rho)\rho_x = 0$.

A generic classical system with momentum conserved: Consider the Hamiltonian dynamics ($\dot{p}_j = -\partial_{q_j} H, \dot{q}_j = \partial_{p_j} H$) generated by the Hamiltonian

$$\sum_{j=1}^{N} p_j^2 + V(q_{j+1} - q_j). \tag{11}$$

(Assume periodic boundary conditions.)

Conserved quantities for generic V: Energy, momentum, particle number.

Hydrodynamic limit: In the large N limit, one must keep track of *densities* corresponding to *each* of the conserved quantities. These densities evolve according to a set of coupled equations. A result in this direction (with small added noise) is [Olla-Varadhan-Yau, 1993].

Integrability of Toda

The Lax matrix: Recall the Flaschka variables $a_j = e^{\frac{1}{2}(q_j - q_{j+1})}, \qquad b_j = p_j.$

Lax matrix (tri-diagonal and symmetric): Let $L = L(t)$ defined by

$$(L)_{jj} = b_j(t), \qquad (L)_{j,j+1} = (L)_{j+1,j} = a_j(t).$$

Lax pair: Along any run of the dynamics,

$$\dot{L} = [B, L],$$

for an explicit skew-symmetric matrix $B(t)$. The Toda lattice is an integrable system (in the classical mechanics sense).

Key consequence:

If periodic b.c.: The trace of any power of $L(t)$ is conserved in time.

In infinite volume, there are infinitely many conserved quantities!

Conserved charges and currents

Charges and currents: Define the *local charges*

$$\mathfrak{t}_j^{[m]}(t) := (L(t)^m)_{jj}, \quad m \geq 0.$$

Claim: At large scales, (local averages of) the observables above vary slowly in time, and "capture" the entire configuration.
Define the corresponding *currents* as

$$\mathfrak{j}_i^{[m]}(t) := a_{i-1}(t)(L(t)^m)_{i,i-1}, \quad m \geq 0.$$

Continuity equation: The Lax equation $\dot{L} = [B, L]$ gives us

$$\frac{d}{dt}\mathfrak{t}_k^{[m]}(t) := \mathfrak{j}_k^{[m]}(t) - \mathfrak{j}_{k+1}^{[m]}(t).$$

After averaging, these equations converge to an infinite collection of (coupled) equations describing the *hydrodynamic limit*; one equation for each $m \geq 0$.

Main observable of interest: Spatial integrated currents

Integrals of charge densities: For $\mathcal{I}$ an interval and t a time, $\sum_{q_j(t)\in\mathcal{I}} \mathfrak{k}_j^{[m]}(t)$. (Jointly across different times.)

Spatial integrated currents: For $q, q' \in \mathbb{R}$, define

$$J_t^{[m]}(q,q') := -\sum_{q_j(t)<q'} \mathfrak{k}_j^{[m]}(t) + \sum_{q_j(0)<q} \mathfrak{k}_j^{[m]}(0)$$

The infinite sums are not well-defined. We use a certain finite-number-of-particles truncation of the model, so that this quantity makes sense. Another representation will be given later in the talk.

Recall interpretation at $m = 0$: The net number of particles crossing the ray from $(q, 0)$ to (q', t) in time t. For $m > 0$, it is the "amount of mth charge" transported along the line segment.

Lax DOS and hydrodynamic limit equations of Toda

Lax density of states: Note: Under thermal equilibrium $\mu_{\beta,\theta;\infty}$, $L(0)$ is a *random* tri-diagonal symmetric matrix. Define the *Lax density of states* ϱ, as

$$\varrho := \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \delta_{\lambda_i}.$$

Above, $\{\lambda_i\}$ are eigenvalues of an $N \times N$ block of the Lax matrix $L(0)$ (under thermal equilibrium). We have $\frac{1}{N}\sum_{k=1}^{N} \mathfrak{k}_k^{[m]} \to \int \lambda^m \varrho(\lambda) d\lambda$.
Stretch: In addition, let

$$\alpha = \mathbb{E}[q_{i+1}(0) - q_i(0)].$$

Main question: How do fluctuations of integrated current scale with time, and what is their distribution? Is it different from ASEP?

Integrated current fluctuations in Toda lattice

Theorem (Aggarwal-N., 2026+)

Let $J^{[m]}_{T\tau}(T\mathfrak{q}, T\mathfrak{q}')$ be the integrated current in the Toda lattice initialized from thermal equilibrium, with the parameter $\theta > 0$ small enough. There are explicit jointly Gaussian processes $\mathfrak{Y}$ and $\mathcal{Z}$ such that the following (joint) convergence in distribution holds. For $t = \tau T$, $q = T\mathfrak{q}$, $q' = T\mathfrak{q}'$, as $T \to \infty$

$$T^{-1/2}\Big(J^{[m]}_{T\tau}(T\mathfrak{q}, T\mathfrak{q}') - \mathbb{E}[J^{[m]}_{T\tau}(T\mathfrak{q}, T\mathfrak{q}')]\Big) \xrightarrow{d} \mathfrak{Y}(\mathfrak{q}, \mathfrak{q}', \tau, m) + \alpha^{-1}\int_{\mathbb{R}} \lambda^m \mathcal{Z}(\lambda, \mathfrak{q}', \tau)\varrho(\lambda)\, d\lambda. \quad (12)$$

Consequence: Integrated currents exhibit *Gaussian fluctuations* at the $T^{1/2}$ scale.

Euler scale fluctuations

- Let $\mathcal{W}$ denote a space-time white noise on $L^2(\mathbb{R}^2, dr\varrho(\lambda)d\lambda)$.
- Let $\mathbf{D}$ denote the *dressing operator* on $L^2(\mathbb{R}, \varrho(\lambda)d\lambda)$.
- If $f \in L^2(\mathbb{R}, \varrho(\lambda)d\lambda)$, let $f^{\mathrm{dr}} := \mathbf{D}f$.
- Let $\varsigma_m(\lambda) := \lambda^m$, $m \geq 0$.
- Let $v_{\mathrm{eff}}(\lambda) := \frac{\varsigma_1^{\mathrm{dr}}}{\varsigma_0^{\mathrm{dr}}}$.

Theorem (Aggarwal-N., 2026+)

For $\theta > 0$ small enough, for any finitely many smooth compactly supported test functions f_i, times $t_i = \tau_i T$, and nonnegative integers m_i

$$T^{-1/2} \sum_{j \in \mathbb{Z}} f_i(j/T)\big(\mathfrak{k}_j^{[m_i]}(t_i) - \mathbb{E}[\mathfrak{k}_j^{[m_i]}(t_i)]\big) \xrightarrow{d} \mathcal{W}\big(f_i(r + \alpha^{-1}\tau_i v_{\mathrm{eff}}(\lambda))\, \varsigma_{m_i}^{\mathrm{dr}}(\lambda)\big).$$

Dressing operator

Scattering shift: Define the kernel

$$T(\lambda, \mu) = 2 \log |\lambda - \mu|.$$

Dressing operator: Let $\mathbf{T}$ be the corresponding integral operator, and for $h \in \mathcal{H} = L^2(\mathbb{R}, \varrho(\lambda) d\lambda)$, define the *dressing operator*

$$h^{\mathrm{dr}}(\lambda) = (1 - \theta \mathbf{T} \boldsymbol{\varrho}_{\boldsymbol{\beta}})^{-1} h.$$

The operator $\boldsymbol{\varrho}_{\boldsymbol{\beta}}$ is a certain multiplication operator.

Processes $\mathfrak{Y}$, $\mathfrak{Z}$

Dressed white noise: Define $\mathcal{W}^{\mathrm{dr}}$ by $\mathcal{W}^{\mathrm{dr}}(f(r,\lambda)) = \mathcal{W}((r,\lambda) \mapsto f_r^{\mathrm{dr}}(\lambda))$.
Gaussian processes: Denote the two families of functions

$$\psi_{\Lambda,\kappa,\tau}(r,\lambda) := 2\log|\Lambda - \lambda| \cdot \left(\mathbb{1}_{\alpha(\kappa-r)>0} - \mathbb{1}_{\alpha(\kappa-r)+\tau(v_{\mathrm{eff}}(\Lambda)-v_{\mathrm{eff}}(\lambda))>0}\right),$$

and

$$\phi^{[m]}_{\mathfrak{q},\mathfrak{q}',\tau}(r,\lambda) := \lambda^m \cdot \left(\mathbb{1}\{\mathfrak{q} < \alpha r\} - \mathbb{1}\{\mathfrak{q}' < \alpha r + \tau v_{\mathrm{eff}}(\lambda)\}\right).$$

Let

$$\mathfrak{X}(\Lambda,\kappa,\tau) := \mathcal{W}^{\mathrm{dr}}(\psi_{\Lambda,\kappa,\tau}), \qquad \mathfrak{Y}(\mathfrak{q},\mathfrak{q}',\tau,m) := \mathcal{W}^{\mathrm{dr}}(\psi_{\Lambda,\kappa,\tau})$$

$$\mathcal{Z}(\Lambda,\mathfrak{q},\tau) := \frac{1}{\varsigma_0^{\mathrm{dr}}(\Lambda)}\left(\Lambda \mapsto \mathfrak{X}(\Lambda, \frac{\mathfrak{q} - \tau v_{\mathrm{eff}}(\Lambda)}{\alpha}, \tau)\right)^{\mathrm{dr}} \tag{13}$$

Charge and current revisited

Effective velocity: $v_{\mathrm{eff}}(\lambda) := \frac{\varsigma_1^{\mathrm{dr}}(\lambda)}{\varsigma_0^{\mathrm{dr}}(\lambda)}$.

Recall current and charge fluctuations: Using the dressing operator, the Gaussian processes defined on the previous slide, the white noise $\mathcal{W}$, and v_{eff},

$$T^{-1/2}\Big(J_{T\tau}^{[m]}(T\mathfrak{q}, T\mathfrak{q}') - \mathbb{E}[J_{T\tau}^{[m]}(T\mathfrak{q}, T\mathfrak{q}')]\Big) \xrightarrow{d} \mathfrak{Y}(\mathfrak{q}, \mathfrak{q}', \tau, m) + \alpha^{-1}\int_{\mathbb{R}} \lambda^m \mathcal{Z}(\lambda, \mathfrak{q}', \tau)\varrho(\lambda)\, d\lambda.$$

$$T^{-1/2}\sum_{j\in\mathbb{Z}} f_i(j/T)\big(\mathfrak{k}_j^{[m_i]}(t_i) - \mathbb{E}[\mathfrak{k}_j^{[m_i]}(t_i)]\big) \xrightarrow{d} \mathcal{W}\big(f_i(r + \alpha^{-1}\tau_i v_{\mathrm{eff}}(\lambda))\,\varsigma_{m_i}^{\mathrm{dr}}(\lambda)\big) \quad (14)$$

Mean-field free energy of high-temperature log gas, link to Toda

Define

$$\mathcal{F}^{\circ}_{\mathrm{de}}(\varrho) = \int_{\mathbb{R}} dw\, \varrho(w) \left(V(w) + \log \varrho(w) + \log P \; - \; \theta \int_{\mathbb{R}} dw'\, \varrho(w') \log |w - w'| \right).$$

- $\mathcal{F}^{\circ}_{\mathrm{de}}$ is the *large deviation functional* of a *high-temperature* β *ensemble*.
- **Lax density of states.** Let ϱ be the limiting eigenvalue measure corresponding to $L(0)$. We have the relation
$$\varrho = \partial_{\theta}(\theta \varrho_{\beta}).$$
- There is a mechanism linking the minimizer ϱ_{β} of $\mathcal{F}^{\circ}_{\mathrm{de}}$ to the limiting Toda eigenvalue density; a link of this form seems to always hold in *generalized hydrodynamics*.

Remark on universality

Dressing operator: Each (say, classical) integrable system has its own scattering kernel T, analog of the densities ϱ_β and ϱ, Hilbert space $\mathcal{H}$, effective velocities $v_{\mathrm{eff}}(\lambda)$.

Claim: This data is all we used to present the scaling limits of current and charges. We expect the form of the answer to be universal, as a function of the data listed above.

Two-point functions

Equilibrium setting: Fix $\beta > 0$ and $\theta < \theta_0(\beta)$, and let $\mu_{\beta,\theta;\infty}$ be the infinite-volume Toda equilibrium measure.

Local conserved fields: For each $m \geq 0$, let $\mathfrak{t}_j^{[m]}(t) = (L(t)^m)_{jj}$ denote the m-th conserved density at site j and time t.

Space–time covariance: For integers $m, n \geq 0$, define

$$S_{m,n}(j,t) := \mathrm{Cov}\big(\mathfrak{t}_j^{[m]}(t),\, \mathfrak{t}_0^{[n]}(0)\big).$$

Operator $\mathbf{F} : \mathcal{H} \to \mathcal{H}$: Define, for $h \in \mathcal{H}$,

$$\mathbf{F}h = h^{\mathrm{dr}} - \Big(\int_{\mathbb{R}} h(\lambda)\varrho(\lambda)d\lambda\Big)\varsigma_0^{\mathrm{dr}}.$$

Goal: Understand the large-scale ($j \sim xT$, $t \sim T$) behavior of these covariances.

Euler-scale limit of two-point functions

Corollary (of the proofs)

For $j = \lfloor xT \rfloor$ and $t = T$, the following asymptotic holds. In the sense of pairing against test functions, for $m, n \geq 1$,

$$S_{m,n}(j, T) \sim \frac{1}{T} \langle \mathbf{F}\varsigma_m, \delta(x - \alpha^{-1}\tau v_{\mathrm{eff}}(\lambda)) \cdot \mathbf{F}\varsigma_n \rangle_{\varrho}$$

For compactly supported, smooth f (and $m, n \geq 1$):

$$\lim_{T\to\infty} \sum_{j\in\mathbb{Z}} f(j/T)\, S_{m,n}(j, \tau T) = \int_{\mathbb{R}} (\mathbf{F}\varsigma_m)(\lambda)\, (\mathbf{F}\varsigma_n)(\lambda)\, f\big(\alpha^{-1}\tau v^{\mathrm{eff}}(\lambda)\big)\, \varrho(\lambda)\, d\lambda.$$

Interpretation: correlations propagate ballistically along characteristics $x = \alpha^{-1}\tau v^{\mathrm{eff}}(\lambda)$.

Related results

Toda generalized hydrodynamics: The Euler equations describing the law-of-large-numbers behavior of the Toda lattice were derived in [Spohn, 2020] and verified rigorously by results of [Mazzucca, 2022], [Guionnet-Mazzucca, 2022], [Mazzucca-Memin, 2024].

Analogous results for hard-rods model: Rods of some length a travel ballistically with some velocity along the real line until they collide, and collisions are elastic. Analogous results: [Ferrari-Franceschini-Grevino-Spohn, 2023], [Ferrari-Olla, 2025].

Box-ball generalized hydrodynamics: Equations describing the *Box-ball system* predicted by generalized hydrodynamics were proven to hold in [Croydon-Sasada, 2021].

Other asymptotic results about Toda lattice: [McLaughlin, 1987], [Venakides-Deift-Oba, 1991], [Deift-Kamvissis-Kriecherbauer-Zhou, 1996], [Bloch-Golse-Paul-Uribe, 2003]

Proof strategy and key tools

Open Toda chain

Index set: Let $N_1 < 0 < N_2$ be integers. We consider the $N = N_2 - N_1 + 1$ particle truncation of the Toda lattice, with phase space $\{(p_j, q_j)_{j \in [N_1, N_2] \cap \mathbb{Z}}\}$.

Open Toda equations of motion:

$$\dot{q}_j = p_j, \qquad \dot{p}_j = e^{q_{j-1} - q_j} - e^{q_j - q_{j+1}}, j = N_1, \ldots, N_2.$$

For all $t \geq 0$, we set $q_{N_1 - 1}(t) = -\infty$ and $q_{N_2 + 1} = +\infty$ by convention.

Lax matrix: The corresponding Lax matrix $L(t)$ is exactly the block of the original infinite Lax matrix corresponding to indices in $[N_1, N_2] \times [N_1, N_2]$. Surprisingly, there is still a Lax pair formulation of the dynamics: for an explicit matrix $Q = Q(t)$,

$$\dot{L} = [Q, L]$$

Conservation of eigenvalues: Therefore, the spectrum of $L(t)$ is independent of t.

Open Toda chain

Open Toda thermal equilibrium :

$$\mu_{\beta,\theta;N}(da\,db) \propto \prod_{j\in[[N_1,N_2-1]]} a_j^{2\theta-1} e^{-\beta a_j^2}\, da_j \cdot \prod_{j\in[[N_1,N_2]]} e^{-\beta b_j^2/2} db_j.$$

Anderson Localization:

Lemma (Kunz-Souillard, 1980, Schenker, 2009)

Under thermal equilibrium, let $(u_j(i))_{i=N_1,\dots,N_2}$ *be the (unit) eigenvector of* $L(t)$ *of eigenvalue* λ_j*. For some index* $\varphi_t(j)$*,*

$$|u_j(i)| \leq Ce^{-c|i-\varphi_t(j)|}.$$

In addition, $\varphi_t(j)$ *is (approximately) unique, and* $j \mapsto \varphi_t(j)$ *can be chosen as a bijection* $[[1,N]] \to [[N_1,N_2]]$*.*

Integrable systems = collections of quasi-particles

Localization center bijection: Let $\varphi_t : [[1, N]] \to [[N_1, N_2]]$ be the *localization center bijection* as defined in the previous slide.

Quasi-particles: Define

$$Q_j(t) = q_{\varphi_t(j)}(t).$$

Quasi-particle evolution: If only a few particles, quasi-particle Q_j travels straight with speed λ_j until hitting other particles, and is shifted by $2 \log |\lambda_j - \lambda_k|$ for each particle Q_k it interacts with.

Main result of [Aggarwal, 2025a]: This holds under thermal equilibrium: On an event of probability at least $1 - c^{-1} e^{-c(\log N)^2}$,

$$Q_i(t) = Q_i(0) + \lambda_i t + \sum_{\varphi_0(j) < \varphi_0(k)} 2 \log |\lambda_j - \lambda_k| - \sum_{\varphi_t(j) < \varphi_t(k)} 2 \log |\lambda_j - \lambda_k| + O((\log N)^C)$$

uniformly in large space-time windows, so long as t is slightly smaller than N.

Quasi-particle effective velocities

Localization center bijection: Let $\varphi_t : [[1, N]] \to [[N_1, N_2]]$ be the *localization center bijection* as defined in the previous slide.

Quasi-particles: Define

$$Q_j(t) = q_{\varphi_t(j)}(t).$$

Define the centered quasi-particle fluctuation:

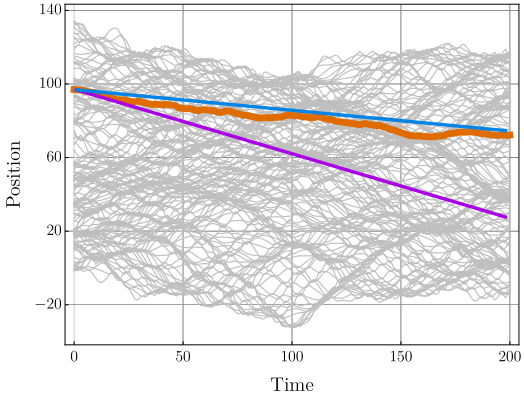
$$Z_i(t) = T^{-1/2}\big(Q_i(t) - Q_i(0) - t v_{\mathrm{eff}}(\lambda_i)\big).$$

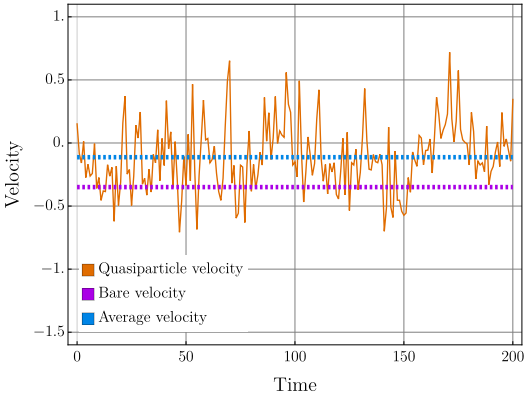
Main result of [Aggarwal, 2025b]: For small enough $\theta > 0$, on an event of probability at least $1 - c^{-1}e^{-c(\log N)^2}$,

$$|Z_i(t)| \leq (\log N)^{35}$$

uniformly in large space-time windows, so long as t is slightly smaller than N.

Quasi-particle effective velocities





We thank Alberto Brollo for this simulation

Quasi-particle fluctuations: main strategy

Let $Q_i(t)$ be the Toda lattice quasi-particle trajectory with spectral label λ_i.

Local charges and currents in terms of quasi-particles: Functionals of local charges can be written in terms of quasi-particles:

$$\sum_{q_j(t)\in\mathcal{I}} \mathfrak{t}_j^{[m]} \approx \sum_{Q_j(t)\in\mathcal{I}} \lambda_j^m \tag{15}$$

In addition, for currents

$$J_t^{[m]}(q,q') \approx -\sum_{j:q'<Q_j(0),Q_j(t)<q} \lambda_j^m + \sum_{j:Q_j(0)<q',q<Q_j(t)} \lambda_j^m \tag{16}$$

Proof strategy: Pin down the exact behavior of $Z_i(t)$, and show a robust convergence theory for linear statistics of random Lax matrices. These two ingredients should allow to extract scaling limits of (16) and (15).

Quasi-particle fluctuations: main probabilistic statement

Let $Q_i(t)$ be the Toda lattice quasi-particles when initialized from thermal equilibrium.

Define the re-indexed quasi-particle fluctuations and re-indexed eigenvalues

$$Z_i^Q(t) = Z_{\varphi_0^{-1}(i)}(t), \qquad \Lambda_i = \lambda_{\varphi_0^{-1}(i)}$$

Theorem (Aggarwal-N., 2026+)

Let $\theta > 0$ be small enough. For T and N large enough, with $T << N$, there exists a coupling such that with probability at least $1 - c^{-1}e^{-c(\log N)^2}$,

$$Z_i^{\mathcal{Q}}(t) \approx \mathcal{Z}(\Lambda_i, \alpha\frac{i}{T} + \frac{t}{T}v_{\mathrm{eff}}(\Lambda_i), \frac{t}{T})$$

uniformly in large space-time windows.

Quasi-particle fluctuations: Brownian motion

Corollary

Let $\theta > 0$ be small enough. Quasi-particle fluctuations $Z_i^{\mathcal{Q}}(t)$ follow the trajectory of a Brownian motion with an explicit covariance.

This verifies predictions from physics literature [Gopalakrishnan-Huse-Khemani-Vasseur, 2018], [De Nardis-Bernard-Doyon, 2019].

Conceptual proof structure

Step 1: Hamiltonian dynamics $\to$ quasi-particles

- express particle motion via spectral data of the Lax matrix,
- linearize around equilibrium to obtain a linear equations for Z_i^Q.

Step 2: couple quasi-particles to Gaussian fields

- prove CLT-type results for spectral linear statistics,
- construct a strong coupling to $\mathcal{Z}$.

Step 3: Use quasi-particles to compute currents and correlations

- rewrite integrated currents as sums over quasi-particles,
- Gaussian fluctuations of quasi-particles + Gaussian fluctuations of linear statistics lead to current fluctuations.